

Narsinha Kailas Kolekar

Software Engineering • AI/ML • Full-Stack Development

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Summary

Full-Stack Software Engineer with hands-on experience building scalable web applications, backend APIs, and machine learning systems using React.js, Node.js, PostgreSQL, and Python-based ML stacks. Passionate about developing data-driven products at the intersection of software engineering and AI/ML, with a strong interest in full-stack development, intelligent systems, and real-world problem solving.

Education

Indian Institute of Information Technology Raichur
B.Tech in Computer Science and Engineering — CGPA: 8.3

Karnataka, India
2024 – 2028

Rajshri Shahu Junior Science College
Class XII: 78.33%

Maharashtra, India
2021 – 2023

Gurukul English School
Class X: 92.8%

Maharashtra, India
2021

Technical Skills

Languages: C, C++, Python, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Streamlit, TensorFlow, Scikit-learn, OpenCV

Databases & Tools: PostgreSQL, MySQL, Git, GitHub, Docker, REST APIs, WebSockets, JWT Authentication, Razorpay API, Jupyter, Google Colab, $\LaTeX$

Core CS / Engineering: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Agile, Scrum, CI/CD, Test Automation

Cloud / Architecture: AWS

Projects

Accident Severity Prediction Using Machine Learning

Python, Scikit-learn, Random Forest, SMOTE, Pandas, NumPy, Streamlit

- Developed an end-to-end ML pipeline using **100K+ accident records**, **60K+ casualty records**, and vehicle datasets to predict accident severity levels.
- Performed data preprocessing, feature engineering, missing value handling, categorical encoding, and SMOTE-based class balancing to improve model performance.
- Benchmarked multiple ML algorithms and achieved **69.9% accuracy** using Random Forest; deployed an interactive Streamlit application for real-time predictions.

Hospital Management System

React.js, Node.js, Express.js, PostgreSQL, REST APIs, WebSockets, JWT Authentication, Razorpay API

- Built secure REST APIs with JWT-based authentication and role-based authorization for managing users, appointments, and healthcare data.
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- Integrated WebSocket-based real-time notifications for appointment updates and doctor-patient communication.
- Implemented Razorpay payment workflows for online consultations, transaction tracking, and automated billing processes.
- Designed scalable database schemas with optimized CRUD operations, validation, and efficient data management using PostgreSQL.

Astra – Women Safety App

Flutter, Firebase, Node.js, Express.js, Real-time Alerts, Location Tracking, Emergency Response

- Built a women safety application with **SOS alerts**, live location sharing, and instant emergency contact notifications for emergency situations.
- Implemented backend APIs and real-time alert workflows for emergency request handling, user profile management, and emergency contact synchronization.
- Engineered a panic-button workflow with location-aware alerts, delivering sub-2-second response times for users.

Automatic Image Colourisation Using Deep Learning

Python, TensorFlow, Keras, OpenCV, CNNs, LAB Colour Space, NumPy

- Developed a CNN-based deep learning model to automatically generate realistic colours from grayscale images using LAB colour space.
- Built efficient preprocessing, training, and inference pipelines using TensorFlow, Keras, and OpenCV for large-scale image datasets.
- Improved image reconstruction quality through normalization, hyperparameter tuning, and architecture experimentation.

Achievements

- Solved over 250 Data Structures Algorithms problems on LeetCode, spanning medium and hard difficulty levels.
- Achieved **Codeforces Rating 985 (Pupil)** and actively participated in rated competitive programming contests.
- Awarded the **Maharashtra State Intermediate Scholarship** for academic excellence.
- Certifications:** Google Gemini AI | Deloitte Cyber Security | Deloitte Technology Job Simulation